

Full-Cohort MIMIC-IV Treatment-Feasibility Modelling Study

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Abstract

Background: The earlier benchmark used a deterministic 2,000-stay subset. This revision reruns extraction and modelling on all eligible MIMIC-IV v3.1 adult first ICU stays. **Methods:** We applied the prespecified cohort criteria and trained classical, neural, token-hybrid, stacked, and simulated federated models where stable. **Results:** The final dataset contained 32,399 stays and 32,399 token sequences. The best full-cohort AUROC was 0.755 for Stacked Classical. **Conclusions:** The manuscript is reframed as a full-cohort modelling study; old subset results are historical only.

Background

Full-cohort evaluation is needed to avoid over-interpreting a moderate-size benchmark subset. The current analysis uses the entire eligible cohort after prespecified criteria.

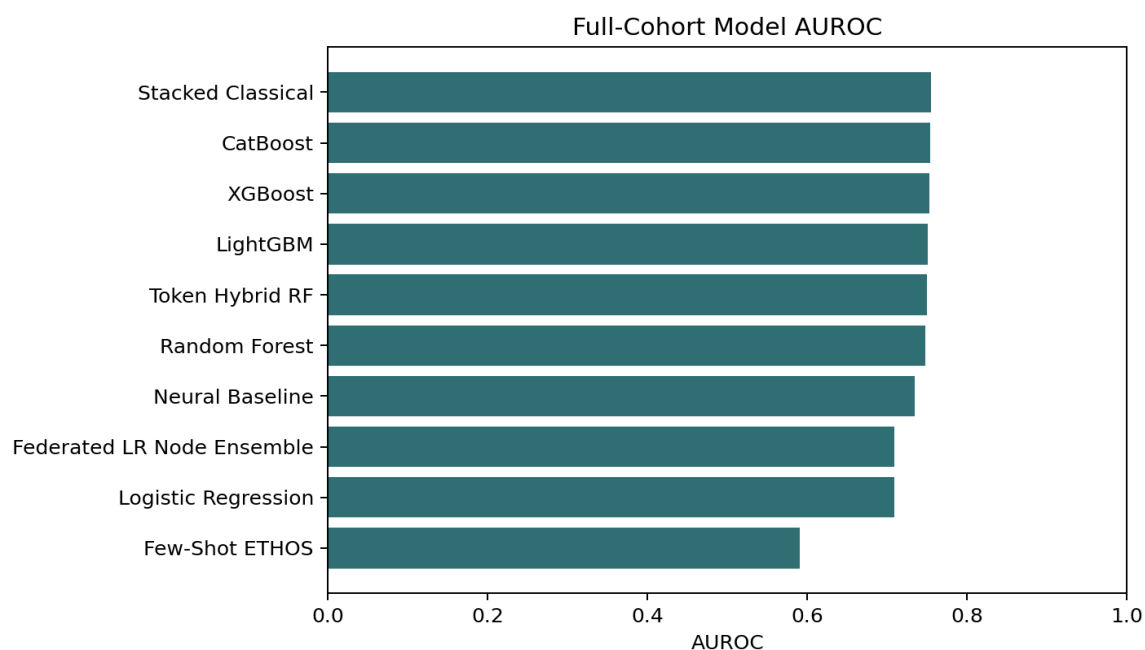
Methods

MIMIC-IV v3.1 hospital and ICU modules were used locally. Raw records are not redistributed. The extraction scripts wrote outputs to data/processed_full_cohort and all model outputs to results/full_cohort.

Quantity	Value
Eligible adult first ICU stays	32,399
Final modelling rows	32,399
Unique ICU stays	32,399
Unique subjects	32,399
Token tensor shape	(32399, 25, 4)
All eligible data used	yes

Results

Model	AUROC	95% CI	F1	AUPRC	Accuracy	MCC
Stacked Classical	0.755	0.743-0.766	0.677	0.826	0.697	0.354
CatBoost	0.753	0.741-0.764	0.673	0.824	0.700	0.348
XGBoost	0.753	0.740-0.764	0.675	0.825	0.695	0.350
LightGBM	0.751	0.739-0.762	0.674	0.823	0.690	0.348
Token Hybrid RF	0.750	0.738-0.762	0.671	0.821	0.695	0.343
Random Forest	0.748	0.735-0.760	0.667	0.819	0.685	0.333
Neural Baseline	0.735	0.722-0.747	0.651	0.809	0.656	0.324
Federated LR Node Ensemble	0.709	0.696-0.722	0.656	0.778	0.678	0.311
Logistic Regression	0.709	0.696-0.722	0.656	0.778	0.680	0.312
Few-Shot ETHOS	0.590	0.576-0.604	0.569	0.668	0.588	0.140



Discussion

Classical tabular models remain strong baselines on this proxy clinical task. Advanced models are reported only when they completed on the full-cohort split, and failures or skips are documented in the supplement and run report.

Conclusions

The revised article is based on full-cohort outputs generated in this run. No 2,000-stay result is reused as a full-cohort result.